

Mingxian Li

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EDUCATION

Sun Yat-sen University

B.Eng. in Microelectronics Science and Engineering

GPA: 3.7/4.0 | Data Structures: 97/100

Sept. 2024 – Present

Shenzhen, China

RESEARCH INTERESTS

Multimodal Reasoning | 3D Spatial Intelligence | AI for Science

RESEARCH EXPERIENCE

HCP Lab, Sun Yat-sen University

Supervised by Xiaodan Liang and Xiao Dong

Mar. 2026 – Sep. 2026

- **SeeChem: 3D Molecular Reasoning in VLMs:** Developed a benchmark with **5,065 questions across 320 molecular instances and five tasks**, and **SeeChem-Data** for verifiable supervision. Built **EasySeeChem-VLM** by integrating **Uni-Mol2** into **Qwen3-VL-8B**; evaluated it alongside **nine external VLMs**.

Manuscript status: Advanced to **Phase 2 review at AAAI 2027**.

HCP Lab, Sun Yat-sen University

Supervised by Xiaodan Liang and Xiao Dong

Oct. 2026 – Present

- **ReactWorld: Geometric World Models for Chemical Reactions:** Exploring non-autoregressive geometric world models for condition-driven reaction product prediction and 3D transition generation.

PROJECT EXPERIENCE

Sun Yat-sen University–Huawei Collaboration

Supervised by Xiaodan Liang and Xiao Dong

Apr. 2026 – Present

- **Chemistry Benchmark and Data Synthesis:** Curated **high-difficulty chemistry problems** from frontier benchmarks and annotated **chain-of-thought solutions** with explicit chemical reasoning for **Huawei's model training**.

PUBLICATIONS

[1] **SeeChem: A Diagnostic Benchmark for 3D Chemical Reasoning in Vision-Language Models** *Under Review*

Mingxian Li^{*}, Kun Xiang^{*}, Xiao Dong^{*}, Haoyuan Li, Zhao Zhen-Hua, Jingbin Tan, Terry Jingchen Zhang, Haiyang Hu, Yangle Fang, Kunkun Liu, Changzheng Zhang, Dandan Tu, Xiaodan Liang

^{*}Equal contribution.

TECHNICAL SKILLS

Programming Languages: Python, C, C++.

Deep Learning Frameworks: PyTorch, TensorFlow.

Development & Computing Tools: Linux, Git/GitHub, Docker, Conda, Slurm, Jupyter Notebook, VS Code.

Writing & Presentation: LaTeX, Markdown, Microsoft Office, draw.io.

SCHOLARSHIPS

- Outstanding Student Scholarship, Sun Yat-sen University 2026
- Special Scholarship, Sun Yat-sen University 2025